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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME1005-1 – FLUID MECHANICS AND MACHINERY
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The specific volume of a liquid is the reciprocal of
 - A) mass density
 - B) weight density
 - C) relative density
 - D) compressibility
2. The value of the surface tension of an ideal fluid is
 - A) zero
 - B) unity
 - C) infinity
 - D) more than that of a real fluid
3. A beaker is filled with a liquid up to the mark of one litre and weighed. The weight of the liquid is found to be 7.5 N. The specific gravity of the liquid will be
 - A) 0.74
 - B) 0.75
 - C) 0.76
 - D) 0.77
4. A liquid jet of 6 cm diameter has a pressure difference of N/m^2 (Take surface tension as 0.075 N/m)
 - A) 2.5
 - B) 5
 - C) 7.5
 - D) 10
5. The pressure _____ as the depth of the liquid increases.
 - A) decreases
 - B) increases
 - C) remains same
 - D) Independent of
6. In venturi meter
 - A) length of converging part is $>$ length of diverging part
 - B) length of converging part is $<$ length of diverging part
 - C) length of converging part is $=$ length of diverging part
 - D) there is no restriction regarding lengths of converging and diverging parts
7. The function of Orifice meter is to measure
 - A) the velocity of the flowing fluid through a pipeline
 - B) the mass flow rate through a pipeline
 - C) the discharge through a pipeline
 - D) friction in the flow through a pipeline
8. The rise in the level of a liquid in a tube is h . What will be the rise in the level if the same amount of liquid is poured into a tube of double the diameter?
 - A) 0
 - B) $h/2$
 - C) h
 - D) $2h$

9. The Euler Turbine equation is based on
- A) Newton's law of viscosity
B) Newton's first law
C) Newton's second law
D) Newton's third law
10. Which of the following statement is False as applied to Degree of Reaction (R)
- A) It is the ratio of Static head to Total head
B) Its value is zero for Impulse machines
C) A machine with any Degree of Reaction other than zero must have an enclosed rotor.
D) For an axial flow machine with 50% reaction, velocity triangles are asymmetrical
11. Which of the following statement is True as applied to Utilization factor (ϵ)
- A) Its value is always more than unity
B) ϵ is minimum for a radial flow machine with radial discharge
C) ϵ is maximum for an axial flow machine with axial discharge
D) it increases with increase in absolute velocity at the exit
12. In an impulse turbine, water at the inlet possesses
- A) only kinetic energy
B) only pressure energy
C) only potential energy
D) all the above
13. Kaplan turbine is preferred as a prime mover where
- A) high head and low discharge are available
B) low head and high discharge are available
C) high head and high discharge are available
D) low head and low discharge are available
14. Hydraulic efficiency is the ratio of
- A) shaft power to input power
B) shaft power to the runner power
C) runner power to the input power
D) runner power to the shaft power
15. Francis turbine is an example of
- A) tangential flow turbine
B) axial flow turbine
C) radial outward flow turbine
D) radial inward flow turbine
16. Propeller turbine with adjustable blades is known as
- A) Pelton turbine
B) Francis turbine
C) Kaplan turbine
D) Curtis turbine
17. Compounding in steam turbines
- A) reduces rotor speed to manageable level
B) reduces energy losses
C) reduces blade erosion
D) all the above
18. In steam turbines, blade efficiency (η_b , nozzle efficiency η_n and gross-stage efficiency η_{g-s}) are related as
- A) $\eta_b = \eta_{g-s} \times \eta_n$
B) $\eta_{g-s} = \eta_b \times \eta_n$
C) $\eta_n = \eta_{g-s} \times \eta_b$
D) $\eta_{g-s} = \eta_b \div \eta_n$
19. If two identical centrifugal pumps each capable of delivering $Q \text{ m}^3/\text{s}$ against head of H meters are connected in parallel the resulting discharge will be
- A) Q against a head of H
B) Q against a head of $2H$
C) $2Q$ against a head of $2H$
D) $2Q$ against a head of H
20. Overall efficiency in centrifugal pumps is always
- A) greater than manometric efficiency
B) lesser than manometric efficiency
C) equal to manometric efficiency
D) greater or lesser than manometric efficiency

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Differentiate the following terms: i) Dynamic viscosity and Kinematic viscosity ii) Adhesiveness and Cohesiveness iii) Mass density and Weight density iv) Compressibility and Bulk modulus	8	L*1	1	1
b)	A horizontal pipe carries oil of specific gravity 0.9. A differential manometer connected at two points 'A' and 'B' of the pipe shows a difference in mercury levels as 25 cm. Find the difference of pressure head in terms of meters water. Also find pressure at point B if the pressure at point A is 2.5 bar.	8	L3	2	2
2. a)	State and prove Pascal's law. Also state Hydrostatic law.	8	L2	1	1
b)	A conical tube is fixed vertically with its small end upwards. Velocity of flow down the tube is 6 m/s at the upper end and 2 m/s at the lower end. Tube is 2.5 m long and the pressure at the upper end is 25 k Pa (abs). Head loss in the tube being 25% of difference in velocity heads at inlet and outlet. Specific gravity of the fluid is 0.85. What is the pressure at the lower end?	8	L3	2	2
3. a)	Derive Euler's equation of motion for ideal fluids and hence deduce Bernoulli's equation.	8	L2	1	1
b)	Derive the equation for discharge through a Venturi meter.	8	L2	2	2
Unit – II					
4. a)	Derive the alternate form of Euler equation and explain the significance of each term.	8	L1	3	1
b)	Show that for maximum utilization and for same absolute velocity, and inlet guide blade angle, the blade speed of 50% reaction turbine is double as that of impulse turbine.	8	L2	4	1
5. a)	Show that for a turbine the maximum utilization factor is given by $C_{max} = \frac{2\phi \cos \alpha_1}{1 + 2\phi R \cos \alpha_1}$ where ϕ = ratio blade velocity at inlet to absolute velocity of fluid at inlet, α_1 = nozzle angle, R = degree of reaction.	8	L2	3	1
b)	The mean rotor blade speed of an axial flow turbine stage with 50% reaction is 225 m/s. Steam emerges from the nozzle inclined at 25° to the plane of the wheel with axial component equal to the blade speed. Assuming symmetric inlet and outlet velocity triangles, determine the rotor blade angles and utilization factor.	8	L3	4	2
6. a)	Derive the condition and expression for maximum Power transfer in case of a Pelton turbine.	6	L2	3	1
b)	A Kaplan turbine working under a head of 20 m develops 11775 k W of shaft power. The outer diameter of runner is 4 m & hub diameter is 2 m. The guide blade angle at the extreme edge of the runner is 25°. The hydraulic and overall efficiencies of the turbines are 90% and 85% respectively. If the velocity of whirl is zero at outlet, determine: i) Runner vane angle at the inlet & outlet at the extreme edge of the runner. ii) Speed of the turbine.	10	L3	4	2
Unit – III					
7. a)	Derive the condition and expression for maximum blade efficiency in case of Parson's turbine.	8	L2	5	1
b)	A De' Laval turbine is used as a prime mover in a thermal power station. The turbine is supplied with a high-pressure steam issuing from the nozzle with a velocity of 900 m/s. The nozzle angle is 21°. Mean blade velocity 350 m/s. The blades are symmetrical. The mass flow rate is 1000 kg/min. Friction factor is 0.9. Determine: i) Blade angles ii) Power developed iii) Blade efficiency iv) Axial thrust.	8	L3	5	2

8. a) Write brief note on following terms as applied to centrifugal pump:

i) Priming ii) Pre-rotation iii) NPSH iv) Cavitation.

8 L1 5 1

- b) A 4 – stage centrifugal pump has impeller each of 40 cm diameter and 2 cm wide at outlet. The outlet vane angle is 45° and the vanes occupy 10% of the outlet area. The manometric efficiency is 90% and overall efficiency is 85%. Determine the head generated by the pump when running at 1000 rpm discharging 60 litres/s. Also determine the power required.

8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
